

Homework 2

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1 Question 1: Passive spread and active propogation on a cable

a) If we take the partial derivatives for $v = e^{-t}u$ we get:

$$v_t = e^{-t}u_t - e^{-t}u \quad (1)$$

$$v_x = e^{-t}u_x \quad (2)$$

$$v_{xx} = e^{-t}u_{xx} \quad (3)$$

$$(4)$$

Substituting this into the original equation:

$$e^{-t}u_t - e^{-t}u = De^{-t}u_{xx} - e^{-t}u \quad (5)$$

$$u_t - u = Du_{xx} - u \quad (6)$$

$$u_t = Du_{xx} \quad (7)$$

Which according to Wikipedia https://en.wikipedia.org/wiki/Heat_equation is the heat equation. A fundamental solution to this equation if we have an impulse at $t=0$ is:

$$u(x, t) = \frac{1}{\sqrt{4\pi Dt}} \exp \left[-\frac{x^2}{4Dt} \right]$$

Which implies

$$v(x, t) = \frac{e^{-t}}{\sqrt{4\pi Dt}} \exp \left[-\frac{x^2}{4Dt} \right]$$

Now taking a bunch of derivatives to substitute it in again. However due to a lot of exponentials, the log is taken to simplify

$$v(x, t) = \frac{e^{-t}}{\sqrt{4\pi Dt}} \exp \left[-\frac{x^2}{4Dt} \right] \quad (8)$$

$$\ln[v] = -t - \ln[\sqrt{4\pi Dt}] - \frac{x^2}{4Dt} \quad (9)$$

$$\frac{v_t}{v} = -1 - \frac{4\pi D}{8\pi Dt} + \frac{x^2}{4Dt^2} \quad (10)$$

$$= v \left(-1 - \frac{1}{2t} + \frac{x^2}{4Dt^2} \right) \quad (11)$$

$$v_t = \frac{e^{-t}}{\sqrt{4\pi Dt}} \exp \left[-\frac{x^2}{4Dt} \right] \left(-1 - \frac{1}{2t} + \frac{x^2}{4Dt^2} \right) \quad (12)$$

$$v_x = \frac{-xe^{-t}}{2Dt\sqrt{4\pi Dt}} \exp \left[-\frac{x^2}{4Dt} \right] \quad (13)$$

$$= \frac{-xv(x, t)}{2Dt} \quad (14)$$

$$v_{xx} = \frac{x^2 e^{-t}}{4D^2 t^2 \sqrt{4\pi Dt}} \exp \left[-\frac{x^2}{4Dt} \right] + \frac{-e^{-t}}{2Dt\sqrt{4\pi Dt}} \exp \left[-\frac{x^2}{4Dt} \right] \quad (15)$$

$$= \frac{x^2 v(x, t)}{4D^2 t^2} + \frac{-v(x, t)}{2Dt} \quad (16)$$

$$(17)$$

Substituting this into the original equation, note I will substitute in a way that keeps $v(x, t)$ as it makes it a lot easier.

$$v \left(-1 - \frac{1}{2t} + \frac{x^2}{4Dt^2} \right) = D \left(\frac{x^2 v(x, t)}{4D^2 t^2} + \frac{-v(x, t)}{2Dt} \right) - v(x, t) \quad (18)$$

$$-1 - \frac{1}{2t} + \frac{x^2}{4Dt^2} = \frac{x^2}{4Dt^2} + \frac{1}{2t} - 1 \quad (19)$$

$$(20)$$

To maximize voltage at some x , we look at the original equation, take a partial with respect to t and then solve for t when the partial equals 0.

$$0 = v(x, t) \left(-1 - \frac{1}{2t} + \frac{x^2}{4Dt^2} \right) \quad (21)$$

There are two ways that the above equation can be solved, either the voltage has to equal 0 or the right term has to equal 0. Looking at the equation for (v, t) , there is no t that forces the equation to equal 0. If we look at the right side however.

$$0 = -1 - \frac{1}{2t} + \frac{x^2}{4Dt^2} \quad (22)$$

$$1 = \frac{x^2 - 2Dt}{4Dt^2} \quad (23)$$

$$4Dt^2 + 2Dt - x^2 = 0 \quad (24)$$

$$(25)$$

Putting it into the quadratic equation:

$$t = \frac{-2D \pm \sqrt{4D^2 + 16Dx^2}}{8D} \quad (26)$$

$$= \frac{-1}{4} \pm \frac{\sqrt{4D^2(1 + \frac{4x^2}{D})}}{8D} \quad (27)$$

$$= \frac{-1}{4} \pm \frac{\sqrt{(1 + \frac{4x^2}{D})}}{4} \quad (28)$$

$$= \frac{1}{4} \left[\pm \sqrt{(1 + \frac{4x^2}{D})} - 1 \right] \quad (29)$$

However we know the impulse occurs at $t = 0$ hence time cannot be negative yielding

$$t_{max} = \frac{1}{4} \left[\sqrt{(1 + \frac{4x^2}{D})} - 1 \right]$$